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DEGRADABLE SUCTION VALVE FOR AN ENDOSCOPE

Abstract

Devices, systems, and methods for a valve assembly for a medical device. The valve has a cap, a shaft which moves within a valve body, and a spring member between the cap and the valve body to move the valve within the body. The valve shaft and/or spring member are made of a degradable material, such as a metal or a polymer. The degradable material has a higher degradation rate than conventional valve components to reduce the environmental impact of disposing of the valve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/613,524 filed on Dec. 21, 2023, the disclosure of which is incorporated herein by reference.

FIELD

[0002] This disclosure relates generally to valve assemblies and methods, and particularly to suction valve assemblies and methods for an endoscope.

BACKGROUND

[0003] A wide variety of intracorporeal medical devices and systems have been developed for medical use, for example, for endoscopic procedures. Some of these devices and systems include guidewires, catheters, catheter systems, endoscopic instruments, and the like. These devices and systems are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. Of the known medical devices, systems, and methods, each has certain advantages and disadvantages.

[0004] Some medical devices include components that are “single use,” intended to be discarded after a short window such as a single day or single procedure. These disposable components contribute significantly to the environmental costs associated with a procedure, as they are often categorized as biological waste. When a component is made of steel or another resilient material, the time and/or energy to break down the component can be significant. There is a need for medical device components that break down more easily to reduce the overall environmental burden associated with their disposal.

SUMMARY

[0005] This disclosure provides design, material, manufacturing method, and use alternatives for medical devices and medical systems. In a first example, a valve assembly for a medical device, includes a valve body having a suction source passage and a working passage; a valve cap positioned above the valve body; a spring member positioned between the valve cap and the valve body such that, when the valve cap is pushed downward relative to the valve body, the spring member applies upward force against the valve cap; and a valve shaft connected to the valve cap, the valve shaft comprising a top opening, a bottom opening, and a side wall having a side opening, the valve shaft configured to translate within the valve body between an upper position and a lower position. The valve stem is made entirely of a degradable material.

[0006] Alternatively or additionally to any of the examples above, the degradable material can be pure Mg or an alloy of Mg with a higher degradation rate than pure Mg.

[0007] Alternatively or additionally to any of the examples above, the degradable material can be pure Zn or an alloy of Zn with a higher degradation rate than pure Zn.

[0008] Alternatively or additionally to any of the examples above, the degradable material can be an alloy of Fe with a higher degradation rate than pure Fe.

[0009] Alternatively or additionally to any of the examples above, the degradable material can be a polymer with a higher degradation rate than polycarbonate.

[0010] Alternatively or additionally to any of the examples above, the polymer can be polylactic acid, polyglycolic acid, polylactic-co-glycolic acid, polycaprolactone, or polyhydroxyalkanoate.

[0011] Alternatively or additionally to any of the examples above, the degradable material can be wood pulp coated with a degradable polymer.

[0012] Alternatively or additionally to any of the examples above, spring member can be made entirely of a degradable material.

[0013] In another example, a valve assembly for a medical device includes a valve body having a suction source passage and a working passage; a valve cap positioned above the valve body; a

spring member made entirely of a degradable material, the spring member positioned between the valve cap and the valve body such that, when the valve cap is pushed downward relative to the valve body, the spring member applies upward force against the valve cap; and a valve shaft connected to the valve cap, the valve shaft comprising a top opening, a bottom opening, and a side wall having a side opening, the valve shaft configured to translate within the valve body between an upper position and a lower position.

[0014] Alternatively or additionally to any of the examples above, the spring member can be a spring lever, a wavy washer stack, or a disc washer stack.

[0015] Alternatively or additionally to any of the examples above, the degradable material can be pure Mg, an alloy of Mg with a higher degradation rate than pure Mg, pure Zn, an alloy of Zn with a higher degradation rate than pure Zn, or an alloy of Fe with a higher degradation rate than pure Fe. Alternatively or additionally to any of the examples above, the degradable material can be a polymer with a higher degradation rate than polycarbonate.

[0016] Alternatively or additionally to any of the examples above, the polymer can be polylactic acid, polyglycolic acid, polylactic-co-glycolic acid, polycaprolactone, or polyhydroxyalkanoate.

[0017] Alternatively or additionally to any of the examples above, the degradable material can be wood pulp coated with a degradable polymer.

[0018] These and other features and advantages of the present disclosure will be readily apparent from the following detailed description, the scope of the claimed invention being set out in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various embodiments and together with the description serve to explain the principles of the present disclosure.

[0020] FIG. 1 depicts a schematic view of components of an illustrative endoscope;

[0021] FIG. 2 depicts a schematic view of components of an illustrative endoscope system;

[0022] FIGS. 3A and 3B depict perspective views of an illustrative suction valve;

[0023] FIG. 3C depicts a schematic view of an illustrative hollow valve shaft of a removable suction valve;

[0024] FIG. 3D depicts a schematic cross-section view of an illustrative suction valve, with the valve in a first configuration;

[0025] FIG. 3E depicts a schematic cross-section view of an illustrative suction valve, with the valve in a second configuration;

[0026] FIG. 3F depicts a schematic cross-section view of an illustrative suction valve, with the valve in a third configuration;

[0027] FIG. 4A depicts a perspective view of an illustrative spring member;

[0028] FIG. 4B depicts a schematic cross-section view of an upper portion of illustrative suction valve with the spring member of FIG. 4A;

[0029] FIG. 5A depicts a perspective view of an illustrative spring member;

[0030] FIG. 5B depicts a schematic cross-section view of an upper portion of an illustrative suction valve with the spring member of FIG. 5A;

[0031] FIG. 6A depicts a perspective view of an illustrative spring member;

[0032] FIG. 6B depicts a schematic cross-section view of an upper portion of an illustrative suction valve with the spring member of FIG. 6A.

[0033] While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It

should be understood, however, that the intention is not to limit the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

[0034] This disclosure is now described with reference to an illustrative medical system that may be used in endoscopic medical procedures. However, it should be noted that reference to this particular procedure is provided only for convenience and not intended to limit the disclosure. A person of ordinary skill in the art would recognize that the concepts underlying the disclosed devices and related methods of use may be utilized in any suitable procedure, medical or otherwise. This disclosure may be understood with reference to the following description and the appended drawings, wherein like elements are referred to with the same reference numerals.

[0035] All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about”, in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (e.g., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.

[0036] The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5). Although some suitable dimensions, ranges, and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges, and/or values may deviate from those expressly disclosed.

[0037] As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise. It is to be noted that in order to facilitate understanding, certain features of the disclosure may be described in the singular, even though those features may be plural or recurring within the disclosed embodiment(s). Each instance of the features may include and/or be encompassed by the singular disclosure(s), unless expressly stated to the contrary. For simplicity and clarity purposes, not all elements of the disclosure are necessarily shown in each figure or discussed in detail below. However, it will be understood that the following discussion may apply equally to any and/or all of the components for which there are more than one, unless explicitly stated to the contrary. Additionally, not all instances of some elements or features may be shown in each figure for clarity.

[0038] It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect the particular feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

[0039] For the purpose of clarity, certain identifying numerical nomenclature (e.g., first, second, third, fourth, etc.) may be used throughout the description and/or claims to name and/or differentiate between various described and/or claimed features. It is to be understood that the

numerical nomenclature is not intended to be limiting and is illustrative only. In some embodiments, alterations of and deviations from previously-used numerical nomenclature may be made in the interest of brevity and clarity. That is, a feature identified as a “first” element may later be referred to as a “second” element, a “third” element, etc. or may be omitted entirely, and/or a different feature may be referred to as the “first” element. The meaning and/or designation in each instance will be apparent to the skilled practitioner.

[0040] The detailed description is intended to illustrate but not limit the disclosure. Those skilled in the art will recognize that the various elements described may be arranged in various combinations and configurations without departing from the scope of the disclosure. The detailed description illustrates example embodiments of the disclosure.

[0041] With reference to FIG. 1, an illustrative endoscope **100** is depicted and FIG. 2 depicts an illustrative endoscope system **200**. The endoscope **100** may include an elongated tube or shaft **100a** that is configured to be inserted into a subject (e.g., a patient).

[0042] A light source **205** of the endoscope system **200** may feed illumination light to a distal portion **100b** of the endoscope **100**. The distal portion **100b** of the endoscope **100** may house an imager (e.g., CCD or CMOS imager) (not shown). The light source **205** (e.g., lamp) may be located in a video processing unit **210** that processes signals input from the imager and outputs processed video signals to a video monitor (not shown) for viewing. The video processing unit **210** may also serve as a component of an air/water feed circuit by housing a pressurizing pump **215**, such as an air feed pump, in the unit **210**.

[0043] The endoscope shaft **100a** may include a distal tip **100c** (e.g., a distal tip unit) provided at the distal portion **100b** of the shaft **100a** and a flexible bending portion **105** proximal to the distal tip **100c**. The flexible bending portion **105** may include an articulation joint (not shown) to assist with steering the distal tip **100c**. On an end face **100d** of the distal tip **100c** of the endoscope **100** is a gas/lens wash nozzle **220** for supplying gas to insufflate the interior of the patient at the treatment area and for supplying water to wash a lens covering the imager. An irrigation opening **225** in the end face **100d** supplies irrigation fluid to the treatment area of the patient. Illumination windows (not shown) that convey illumination light to the treatment area, and an opening **230** to a working channel **235** extending along the shaft **100a** for passing tools to the treatment area, may also be included on the face **100d** of the distal tip **100c**. The working channel **235** may extend along the shaft **100a** to a proximal channel opening **110** positioned distal to an operating handle **115** (e.g., a proximal handle) of the endoscope **100**. A biopsy valve **120** may be utilized to seal the channel opening **110** against unwanted fluid egress.

[0044] The operating handle **115** may be provided with knobs **125** for providing remote 4-way steering of the distal tip via wires connected to the articulation joint in the bendable flexible portion **105** (e.g., one knob controls up-down steering and another knob control for left-right steering). A plurality of video switches **130** for remotely operating the video processing unit **210** may be arranged on a proximal end side of the handle **115**.

[0045] The handle **115** may be provided with dual valve locations **135**. One of the valve locations **135** may receive a gas/water valve **140** for operating an insufflating gas and lens water feed operation. A gas supply line **240a** and a lens wash supply line **245a** run distally from the gas/water valve **140** along the shaft **100a** and converge at the distal tip **100c** proximal to the gas/wash nozzle **220** (FIG. 2).

[0046] The other valve location **135** may receive a suction valve **145** for operating a suction operation. A suction supply line **250a** may run distally from the suction valve **145** along the shaft **100a** to a junction point in fluid communication with the working channel **235** of the endoscope **100**.

[0047] The operating handle **115** may be electrically and fluidly connected to the video processing unit **210**, via a flexible umbilical **260** and connector portion **265** extending therebetween. The flexible umbilical **260** has a gas (e.g., air or CO₂) feed line **240b**, a lens wash feed line **245b**, a

suction feed line **250b**, an irrigation feed line **255b**, a light guide (not shown), and an electrical signal cable (not shown). The connector portion **265** when plugged into the video processing unit **210** connects the light source **205** in the video processing unit with the light guide. The light guide runs along the umbilical **260** and the length of the endoscope shaft **100a** to transmit light to the distal tip **100c** of the endoscope **100**. The connector portion **265** when plugged into the video processing unit **210** also connects the air pump **215** to the gas feed line **240b** in the umbilical **260**. [0048] A water reservoir or container **270** (e.g., water bottle) may be fluidly connected to the endoscope **100** through the connector portion **265** and the umbilical **260**. A length of gas supply tubing **240c** passes from one end positioned in an air gap **275** between the top **280** (e.g., bottle cap) of the reservoir **270** and the remaining water **285** in the reservoir to a detachable gas/lens wash connection **290** on the outside of the connector portion **265**. The gas feed line **240b** from the umbilical **260** branches in the connector portion **265** to fluidly communicate with the gas supply tubing **240c** at the detachable gas/lens wash connection **290**, as well as the air pump **215**. A length of lens wash tubing **245c**, with one end positioned at the bottom of the reservoir **270**, may pass through the top **280** of the reservoir **270** to the same detachable connection **290** as the gas supply tubing **240c** on the connector portion **265**. In other embodiments, the connections may be separate and/or separated from each other. The connector portion **265** may also have a detachable irrigation connection **293** for irrigation supply tubing (not shown) running from a source of irrigation water (not shown) to the irrigation feed line **255b** in the umbilical **260**. In some embodiments, irrigation water is supplied via a pump (e.g., peristaltic pump) from a water source independent (not shown) from the water reservoir **270**. In other embodiments, the irrigation supply tubing and lens wash tubing **245c** may source water from the same reservoir. The connector portion **265** may also include a detachable suction connection **295** for suction feed line **250b** and suction supply line **250a** fluidly connecting a vacuum source (e.g., hospital house suction) (not shown) to the umbilical **260** and endoscope **100**.

[0049] The gas feed line **240b** and lens wash feed line **245b** may be fluidly connected to the valve location **135** for the gas/water valve **140** and configured such that operation of the gas/water valve in the well controls supply of gas or lens wash to the distal tip **100c** of the endoscope **100**. The suction feed line **250b** is fluidly connected to the valve location **135** for the suction valve **145** and configured such that operation of the suction valve **145** in the well controls suction applied to the working channel **235** of the endoscope **100**.

[0050] An example of a removable suction valve **300** is illustrated in FIGS. 3A-3F. The valve cap **302** includes an air escape hole **304** and a spring member **306**. The hollow valve shaft **308** includes open ends **310** and **312**, guide slots **314**, and side openings **316**.

[0051] The valve **300** is inserted into a body **330**, such as one of the locations **135** described above and illustrated in FIGS. 1 and 2. The body **330** is sized and shaped to receive the shaft **308** of the valve **300**, as well as alternative valve designs (including each of those illustrated and described below). The valve body **330** includes a suction source passage **332** communicating with a source of suction as described above with respect to suction supply line **250a**. A working passage **334** communicates with the with the working channel **235** of the endoscope **100** as described above.

[0052] FIG. 3D shows an open configuration for the valve when the air escape hole **304** is unblocked, air passing through valve shaft **308** from the top end **310** to the bottom end **312** and into the suction source passage **332**.

[0053] FIG. 3E shows a closed configuration in which the valve **300** is positioned lower in the body **330**. Downward force on the valve cap **302** causes and maintains this position; when the valve cap **302** is released, the spring member **306** returns the valve **300** to its previous position. The user's finger also blocks the air escape hole **304** in some embodiments; in some implementations, a flap or other obstruction is instead provided to block the air escape hole **304**. In this configuration, the side opening **316** in the hollow valve shaft **308** is aligned with the working passage **334** of the valve body **330**. The working channel **235** of the endoscope **100** is therefore subject to vacuum

from the suction supply line **250a**. Air passes through the working passage **334**, through the side opening **316** into the hollow body of the valve shaft **308**, down through the open bottom end **312**, and into the suction source passage **332**. Upon release of the downward force and return of the valve cap **302** to its previous position, the placement of the lower seal **320c** again prevents additional water from entering the feed through the annular passage.

[0054] FIG. **3F** shows the upper portion of the valve **300**, in which the cap **302** interfaces with the valve body **330** by means of the spring member **306**. The spring member **306** as shown is a coiled spring of steel wire that pushes against the valve body **330** to lift the valve **300** into the upper position.

[0055] The valve shaft **308** may couple to the cap **302** in any suitable manner. In some cases, a portion (e.g., a proximal portion) of the valve stem **308** extending proximally of the air hole **304** may be coupled to the cap **302** via one or more suitable coupling mechanisms. Example suitable coupling mechanisms include, but are not limited to, adhesives, a threaded connection, a luer lock connection, a snap connection, a ball-detent connector, a friction fit, and/or additional or alternative coupling mechanisms.

[0056] The valve shaft **308** may have any suitable configuration configured to adjust positions within the valve well, adjust flow paths to the air and water supplies and feeds, and couple to the cap **302**. The valve **300** may be formed in any suitable manner. In some cases, though not required, the valve **300** may be formed using a molding process, an injection molding process, a casting process, a finishing process, sanding, and/or by or with one or more additional or alternative manufacturing techniques. In one illustrative example, the valve **300** may be formed using an injection molding process.

[0057] In this specification, we define “degradation” as physical and/or chemical changes to a component that occur over time due to the component's exposure to an environment. Weathering, corrosion, and decomposition are three common examples of degradation.

[0058] Steel and aluminum are often chosen as materials for mechanical devices because of the metals' longevity and resistance to degradation. Moreover, particular alloys of steel and/or aluminum are often favored over other alloys due to superior resistance to degradation.

[0059] As an alternative, however, metals can be chosen to specifically have a higher degradation rate—that is, the metals will degrade in the same environment in a shorter period of time. Magnesium, zinc, and iron alloys are known to break down more quickly and have therefore been used as biodegradable materials for temporary insertion into human bodies. Their higher degradation rates also make alloys of these metals good candidates for reducing the environmental impact of disposable components. Both metal and non-metal materials that have a higher degradation rate are “degradable” materials.

[0060] The degradation properties of Mg, Zn, and Fe, as well as alloys of each of these, have been examined. A discussion of many of these materials can be found in Li et al, “Progress of biodegradable materials,” Progress in Natural Science: materials international 24 (2014) 414-422, which is herein incorporated by reference in its entirety. Other alloys are known to those skilled in the art.

[0061] With respect to magnesium, the degradation rates of pure magnesium and alloys such as Mg—Zn—Mn, Mg—Ca, Mg—Sr, Mg—Si, Mg—Zr, AZ91D, AZ31, LAE442, and WE43 may make some or all of them appropriate substitutes for slower-degrading materials, along with other alloys currently known in the art, or later-discovered alloys having similar properties.

[0062] With respect to iron, the degradation rates of alloys such as Fe-3C, Fe-3S, Fe-3W, Fe-10Mn, Fe-10Mn-1Pd, Fe-30Mn(forged), Fe30Mn(cast), Fe-30Mn-1C, and Fe-30Mn-6Si may make some or all of them appropriate substitutes for slower-degrading materials, along with other alloys currently known in the art, or later-discovered alloys having similar properties.

[0063] With respect to zinc, the degradation rates of pure zinc and alloys such as Zn—Mg, Zn—Mg—Ca, Zn—Mg—Sr, Zn—Al, Zn—Mn, Zn—Ca, Zn—Sr, and Zn—Ag may make some or all of

them appropriate substitutes for slower-degrading materials, along with other alloys currently known in the art, or later-discovered alloys having similar properties.

[0064] In addition to metal, a range of degradable polymers have been explored, both for their reduced environmental impact and for their biocompatibility in medical applications. Such plastics have been described in Rosenboom et al., “Bioplastics for a circular economy,” *Nat Rev Mater* 7, 117-137 (2022), which is hereby incorporated by reference in its entirety. Other materials are known to those skilled in the art.

[0065] The degradation rates of polymers such as polylactic acid (PLA), polyglycolic acid (PGA), polylactic-co-glycolic acid (PLGA), polycaprolactone (PCL), and polyhydroxyalkanoates (PHAs) may make them appropriate substitutes for traditional polycarbonate plastics and other rigid materials.

[0066] Returning to the valve **300** described and shown above, in some implementations, the cap **302** may be made of a degradable elastomer, such PCL copolymerized with PGA and PLA. Non-degradable rubber, such as silicon polymers, are currently used for such cap structures. A degradable polymer having similar elastic properties to currently used materials can be selected to reduce the environmental impact of this component.

[0067] The valve shaft **308** may be made of a degradable polymer having tensile strength similar to that of polycarbonate, such as PHA or PLA. Alternatively, the valve shaft may be made of a thin rolled film of a lightweight material and coated with a more waterproof or biocompatible material. For example, in some embodiments, a hollow valve shaft **308** could be formed from paper or pressed pump. A sprayed or dipped coating of a degradable polymer could be added before the film is rolled and then glued, stitched or melted to form the tube shape of the hollow valve shaft **308**. A degradable polymer film could be formed in the same manner. Multiple film layers could be selected for different properties and pressed together to form the film used for the valve shaft **308**.

[0068] In some implementations, the spring member **306** may be made of a degradable material, which may be a degradable metal or a degradable polymer as described above. In some implementations, the degradable spring member may have a significantly lower strength and/or modulus than steel wire; therefore, a different geometry for the member may be used.

[0069] Degradable spring levers, wave washer stacks, and disc washer stacks described. These spring members may be used with a degradable material that is less ductile than steel and more difficult to manufacture as a coil spring, or where a coil spring or similar structure made of the degradable material would not have a sufficient spring constant to allow for easy operation of the valve. For example, a coil spring made of degradable polymer may not have a spring constant sufficient to consistently move the valve as required, while a degradable polymer spring lever provides sufficient force. As another example, a degradable iron alloy might not be sufficiently ductile for standard techniques to form it into a coil spring, but the degradable iron alloy may be efficiently stapled into a disc washer. Each spring member may be molded, stamped, tooled, drawn, rolled, cut and/or any manufacturing technique or combination techniques appropriate for the material and structure.

[0070] FIGS. **4A** and **4B** show a degradable spring member in the form of a spring lever **406**. A plurality of spring levers **406** may be included around the circumference of the cap **302**. As with other versions of the spring member, the spring levers **406** may be attached or affixed to the cap **302** in various manners known in the art. The spring levers **406** may flex when the valve **300** is moved downward against the valve body **330**, pushing upward to restore the valve **300** to the upper position. The spring levers **406** may be made of a degradable polymer such as thermoplastic with a high degradation rate. The spring levers **406** may also be made of a magnesium, zinc, or iron alloy, or any other degradable metal.

[0071] FIGS. **5A** and **5B** show a degradable spring member in the form of a wave washer stack **506**. The wave washer stack **506** may be made of a plurality of wave washers of similar shape, but with the upper and lower portions of the waves offset to contact and provide the basis for

restorative force when the stack is compressed. Pressing downward on the valve cap **302** compresses the wave washer stack **506**, which exerts upward force on the cap **302** when released to move the valve **300** back into the upper position. The wave washer stack **506** may be made of a degradable polymer with a higher degradation rate. The stack **506** may also be made of a magnesium, zinc, or iron alloy, or any other degradable metal.

[0072] FIGS. **6A** and **6B** show a degradable spring member in the form of a disc washer stack **606**. The disc washer stack **606** may be made of a plurality of disc washers, each of which is similarly shaped as a truncated cone or a truncated sphere (sometimes referred to as “Belleville” washers). The stacked disc washers provide a combined spring force that is approximately linear, a sum of the individual spring forces of the washers. Pressing downward on the valve cap **302** compresses the disc washer stack **606**, which exerts upward force on the cap **302** when released to move the valve **300** back into the upper position. The disc washer stack **606** may be made of a degradable polymer with a higher degradation rate. The stack **606** may also be made of a magnesium, zinc, or iron alloy, or any other degradable metal. The overall diameter of the washers, as well as the distance that the inner diameter of each washer is raised to form the truncated conical shape, and the corresponding angle above the horizontal of each washer, can be chosen according to the properties of the degradable material and the requirements of the resulting valve.

[0073] It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A valve assembly for a medical device, comprising: a valve body having a suction source passage and a working passage; a valve cap positioned above the valve body; a spring member positioned between the valve cap and the valve body such that, when the valve cap is pushed downward relative to the valve body, the spring member applies upward force against the valve cap; and a valve shaft connected to the valve cap, the valve shaft comprising a top opening, a bottom opening, and a side wall having a side opening, the valve shaft configured to translate within the valve body between an upper position and a lower position, the valve stem made entirely of a degradable material.
2. The valve assembly of claim 1, wherein the degradable material is pure Mg or an alloy of Mg with a higher degradation rate than pure Mg.
3. The valve assembly of claim 1, wherein the degradable material is pure Zn or an alloy of Zn with a higher degradation rate than pure Zn.
4. The valve assembly of claim 1, wherein the degradable material is an alloy of Fe with a higher degradation rate than pure Fe.
5. The valve assembly of claim 1, wherein the degradable material is a polymer with a higher degradation rate than polycarbonate.
6. The valve assembly of claim 5, wherein the polymer is polylactic acid, polyglycolic acid, polylactic-co-glycolic acid, polycaprolactone, or polyhydroxyalkanoate.
7. The valve assembly of claim 1, wherein the degradable material is wood pulp coated with a degradable polymer.
8. The valve assembly of claim 1, wherein the spring member is made entirely of a degradable material.
9. A valve assembly for a medical device, comprising: a valve body having a suction source passage and a working passage; a valve cap positioned above the valve body; a spring member made entirely of a degradable material, the spring member positioned between the valve cap and

the valve body such that, when the valve cap is pushed downward relative to the valve body, the spring member applies upward force against the valve cap; and a valve shaft connected to the valve cap, the valve shaft comprising a top opening, a bottom opening, and a side wall having a side opening, the valve shaft configured to translate within the valve body between an upper position and a lower position.

10. The valve assembly of claim 9, wherein the spring member is a spring lever.

11. The valve assembly of claim 9, wherein the spring member is a wavy washer stack or a disc washer stack.

12. The valve assembly of claim 9, wherein the degradable material is pure Mg, an alloy of Mg with a higher degradation rate than pure Mg, pure Zn, an alloy of Zn with a higher degradation rate than pure Zn, or an alloy of Fe with a higher degradation rate than pure Fe.

13. The valve assembly of claim 9, wherein the degradable material is a polymer with a higher degradation rate than polycarbonate.

14. The valve assembly of claim 13, wherein the polymer is polylactic acid, polyglycolic acid, polylactic-co-glycolic acid, polycaprolactone, or polyhydroxyalkanoate.

15. The valve assembly of claim 9, wherein the degradable material is wood pulp coated with a degradable polymer.

16. The valve assembly of claim 9, wherein the valve shaft is made entirely of a degradable material.

17. The valve assembly of claim 16, wherein the valve shaft and the spring member are made of the same degradable material.

18. The valve assembly of claim 16, wherein the spring member is made of a first degradable material, and wherein the valve shaft is made of a second degradable material of higher rigidity than the first degradable material.

19. The valve assembly of claim 16, wherein the valve shaft is made entirely of a polymer with a higher degradation rate than polycarbonate.

20. The valve assembly of claim 19, wherein the polymer is polylactic acid, polyglycolic acid, polylactic-co-glycolic acid, polycaprolactone, or polyhydroxyalkanoate.
